VISVESVARAYA TECHNOLOGICAL UNIVERSITY
"Jnana Sangama", Belgaum -590014, Karnataka.

PROJECT WORK-2 REPORT
On

"Restaurant Site (URBAN BITES)"

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Under the Guidance of

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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BENGALURU-560019

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CERTIFICATE

This is to certify that the project work entitled **"RESTAURANT SITE"** carried out by SHREYAS SINHA (1BM23CS321), SARIM ALI (1BM23CS304), SHONAL MAYANK (1BM23CS316), SANCHIT KASHYAP (1BM23CS298) who are bonafide students of **B. M. S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visveswaraiah Technological University, Belgaum during the year 2024. The project report has been approved as it satisfies the academic requirements in respect of **Full Stack Web Development(23CS3AEFWD)** work prescribed for the said degree.

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DECALARATION

We, **SHREYAS SINHA (1BM23CS321), SARIM ALI (1BM23CS304), SHONAL MAYANK (1BM23CS316), SANCHIT KASHYAP (1BM23CS298)** students of 3rd Semester, B.E, Department of Computer Science and Engineering, BMS College of Engineering, Bangalore, hereby declare that, this Full Stack Web Development entitled **" RESTAURANT SITE "** has been carried out by us under the guidance of **Srushti C S, Assistant Professor,** Department of CSE, BMS College of Engineering, Bangalore during the academic semester Dec 2023 - Mar 2024.

We also declare that to the best of our knowledge and belief, the development reported here is not from part of any other report by any other students.

Signature

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ABSTRACT

The Urban Bites Web Application is a comprehensive digital solution developed to redefine and enhance the interaction between customers and restaurants. This project is motivated by the increasing demand for convenience and efficiency in modern dining experiences. The application serves as a seamless platform that empowers customers to explore menus, place reservations, and order food for delivery—all from their personal devices, thereby simplifying and enriching their dining journey.

The core objective of the Urban Bites Web Application is to maximize user convenience by addressing common pain points in the dining process. The platform enables users to browse detailed menus effortlessly, make informed dining decisions, and reserve tables with just a few clicks. Additionally, the home delivery feature caters to the growing preference for enjoying restaurant-quality meals at home. By streamlining these processes, the application reduces complexity and fosters a user-centric dining experience.

For restaurant owners, the application offers robust operational management capabilities. Powered by a MongoDB backend, the system centralizes data management, enabling streamlined handling of reservations, menu updates, and delivery orders. This centralized approach not only reduces manual workload but also enhances real-time tracking and operational efficiency. The automation of these processes allows restaurant staff to focus more on delivering exceptional service and improving customer satisfaction.

The Urban Bites Web Application prioritizes user experience through its clean, intuitive, and responsive design. Modern web technologies ensure that navigation is straightforward, providing customers with an enjoyable and frictionless interaction. Whether users are browsing the menu, booking a table, or ordering delivery, the platform's user-friendly interface fosters loyalty and encourages repeat engagement.

In summary, the Urban Bites Web Application bridges the gap between customers' expectations and restaurant operations. By integrating convenience, efficiency, and user-centric design, the application transforms the dining experience into one that is effortless and accessible. This innovative solution not only benefits customers but also empowers restaurant owners with tools to streamline their operations, ultimately contributing to a more satisfying and profitable dining ecosystem.

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1. INTRODUCTION

1.1 Overview:

The Urban Bites Full-Stack Web Application is a modern and feature-rich solution for managing a restaurant's operations. The application aims to streamline core activities such as menu display, table reservations, and home delivery orders, providing an optimized experience for both customers and restaurant administrators.

Technological Stack:

- **Frontend:** Developed using **HTML, CSS, and JavaScript** to create a dynamic and interactive user interface. Focused on creating visually appealing layouts to enhance user engagement and satisfaction.

- **Backend:** Built with **Node.js** for efficient server-side scripting and event-driven architecture. Utilized **Express.js**, a lightweight framework, to simplify routing and backend logic for handling requests like reservations, menu updates, and order management.

- **Database:** Employed **MongoDB** for its flexibility in handling hierarchical and unstructured data. Integrated robust mechanisms to ensure secure storage and swift retrieval of customer data, reservations, and order details.

Interactive User Experience:

Menu Display: Dynamic and categorized menu sections that allow customers to browse dishes effortlessly.

Search Functionality: Enable customers to search for dishes by name or category.

Order Tracking: Provide live updates on food delivery status for a transparent experience.

Dynamic Backend:
Menu management tools for administrators to update items, prices, and availability in real time.
Automated reservation handling that prevents double-booking of tables.
Efficient order management system to prioritize and dispatch delivery requests seamlessly.

Efficient Data Management:
Ensured robust data security with MongoDB's authentication mechanisms.
Optimized data retrieval for quick loading times, even under high traffic.
Utilized indexing and schema validation to maintain data integrity.

Seamless Integration:
Integrated the frontend and backend using **RESTful APIs** to ensure smooth data exchange.
Implemented API endpoints for creating, reading, updating, and deleting resources like menu items, reservations, and orders.
Enhanced communication between modules to provide real-time updates on reservations and orders.

Scalability:
Designed with scalability in mind to accommodate future growth, such as an increasing number of users or expanding restaurant chains.
Incorporated load-balancing techniques to distribute server traffic efficiently.
Leveraged the flexibility of MongoDB's document-oriented architecture to scale horizontally when needed.

Additional Features for Enhanced Functionality:

User Authentication: Secure login and account creation for customers and staff using JWT (JSON Web Tokens).

Admin Dashboard: Tools for restaurant administrators to analyze sales, manage menu items, and monitor reservations.

Payment Integration: Support for online payment gateways for seamless and secure transactions.

Key Objectives and Outcomes:

The Urban Bites Web Application is designed to:
Enhance customer satisfaction by providing a modern and intuitive platform for dining and delivery services.
Reduce operational bottlenecks by automating table reservations and order processing.
Offer data-driven insights to restaurant administrators for informed decision-making.
Scale effortlessly to meet growing demands, ensuring the restaurant remains competitive in the evolving food industry landscape.
By merging innovative technologies with practical restaurant management needs, Urban Bites is a transformative tool for modern dining establishments.

1.2 Motivation:

The Urban Bites Web Application was developed with the primary motivation of simplifying and enhancing the interaction between customers and the restaurant. In an age where convenience and efficiency are highly valued, this application aims to provide a seamless platform for customers to explore the restaurant's offerings, place reservations, and order food for delivery—all from the comfort of their own devices. The ultimate goal is to create an effortless user experience that reduces the complexity of these tasks and makes them more accessible.

One of the core motivations behind this project is to improve user convenience. Modern consumers

prefer fast, easy-to-use platforms to meet their needs, and the Urban Bites application directly addresses this by offering a range of functionalities that make the dining process more straightforward. Customers can easily browse the restaurant's menu, allowing them to make informed choices. They can also quickly reserve a table through the website, eliminating the need for phone calls and reducing wait times. Additionally, the ability to place home delivery orders online adds another layer of convenience, catering to customers who prefer enjoying meals from the comfort of their homes.

For restaurant owners, managing customer reservations, food orders, and menu updates can be a complex and time-consuming task. The automation of restaurant operations through this web application was a key motivation. By centralizing data management in a backend powered by MongoDB, the application simplifies the process of managing reservations, menu items, and home delivery orders. This centralization makes it easier for restaurant staff to keep track of orders, view real-time reservation updates, and modify the menu as needed. Moreover, the system allows for quick order processing, which enhances the restaurant’s overall operational efficiency and improves service speed.

Lastly, the application was designed with a strong focus on providing a seamless user experience. The use of modern web technologies ensures that the interface is clean, intuitive, and responsive. Customers can easily navigate the site and interact with its features without confusion. Whether they are browsing the menu, making a reservation, or placing an order, the application is designed to be as user-friendly as possible. This smooth and enjoyable experience not only boosts customer satisfaction but also encourages repeat business, fostering customer loyalty in the long run.

In conclusion, the Urban Bites Web Application was motivated by the desire to enhance both the customer experience and business operations through a simple, efficient, and automated system. The platform provides a one-stop solution for customers and restaurant owners alike, improving the overall dining experience and streamlining restaurant operations.

2. SOFTWARE & HARDWARE REQUIREMENTS

2.1Hardware Requirements:

For the Urban Bites Web Application, the basic hardware requirements are as follows:

- A PC with the following or greater specifications:
- Processor: Intel Core i3 or higher
- RAM: 8 GB or more
- Storage: 500 GB Hard Drive or SSD
- A stable internet connection (2 Mbps or higher)

2.2 Software Requirements:

Operating System: Windows

- Frontend Technologies:
  HTML: For structuring the web pages.
  CSS: For styling and responsive design.
  JavaScript: For client-side functionality and interactivity.

- Backend Technologies:
  Node.js: For building the backend server and handling HTTP requests.
  Express.js: A Node.js framework for managing APIs and routing.
  MongoDB: NoSQL database for storing data related to menu items, reservations, and delivery orders.

- Integrated Development Environment (IDE):
  VS Code: Preferred code editor with support for extensions like ESLint and Prettier for maintaining code quality.

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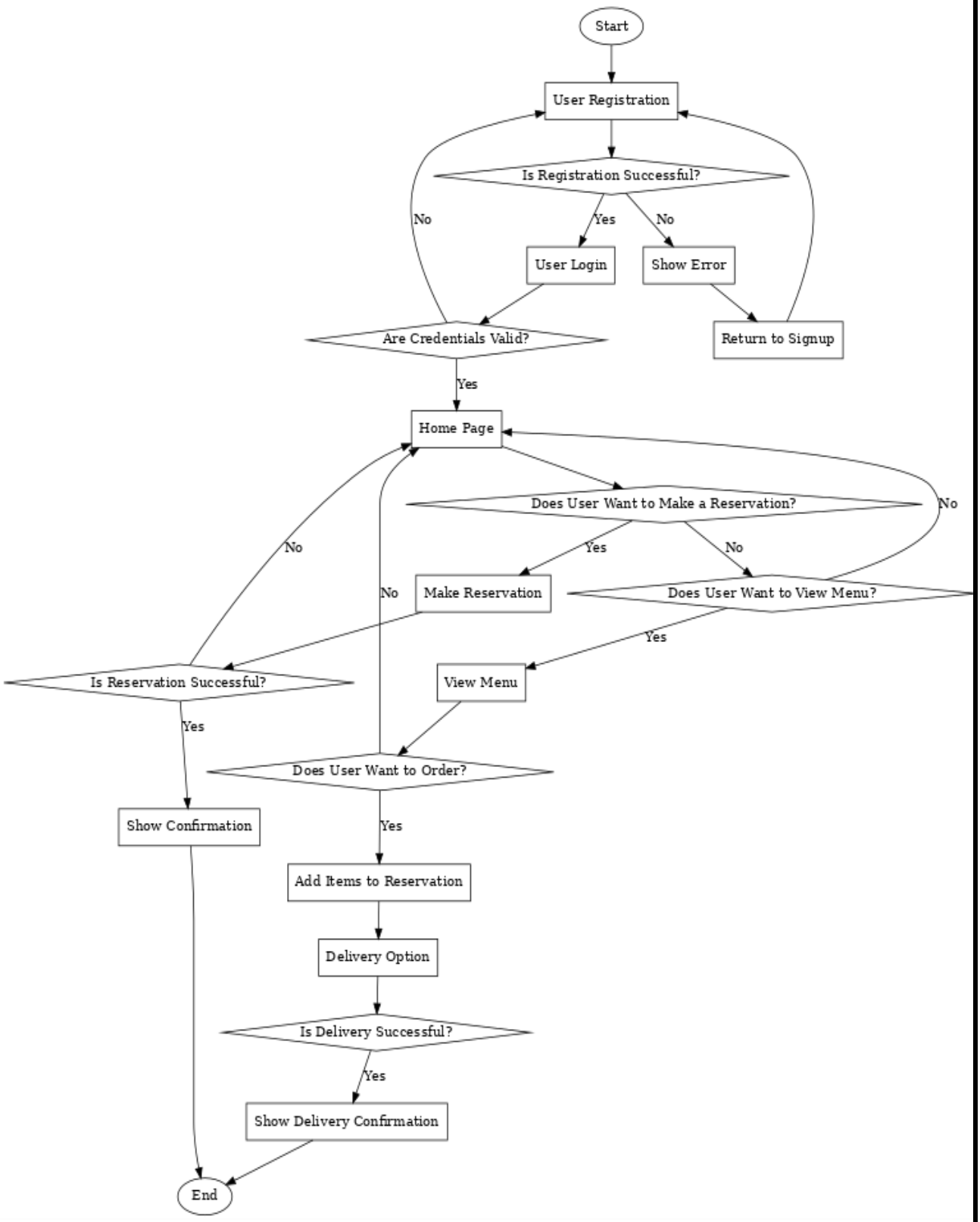
3. ER DIAGRAM OF THE PROJECT



4. USER INTERFACE DESIGN

The Urban Bites Web Application prioritizes user experience through its clean, intuitive, and responsive design. Modern web technologies ensure that navigation is straightforward, providing customers with an enjoyable and frictionless interaction. Whether users are browsing the menu, booking a table, or ordering delivery, the platform's user-friendly interface fosters loyalty and encourages repeat engagement.

LET'S DIVE INTO THE WORLD OF FOOD CREATED BY US.

Login Page The login page provides a gateway for users to access their personalized experience. New users can create an account with a unique username, while returning users can log in seamlessly to proceed to the main page. This ensures secure and convenient access to the application's features.

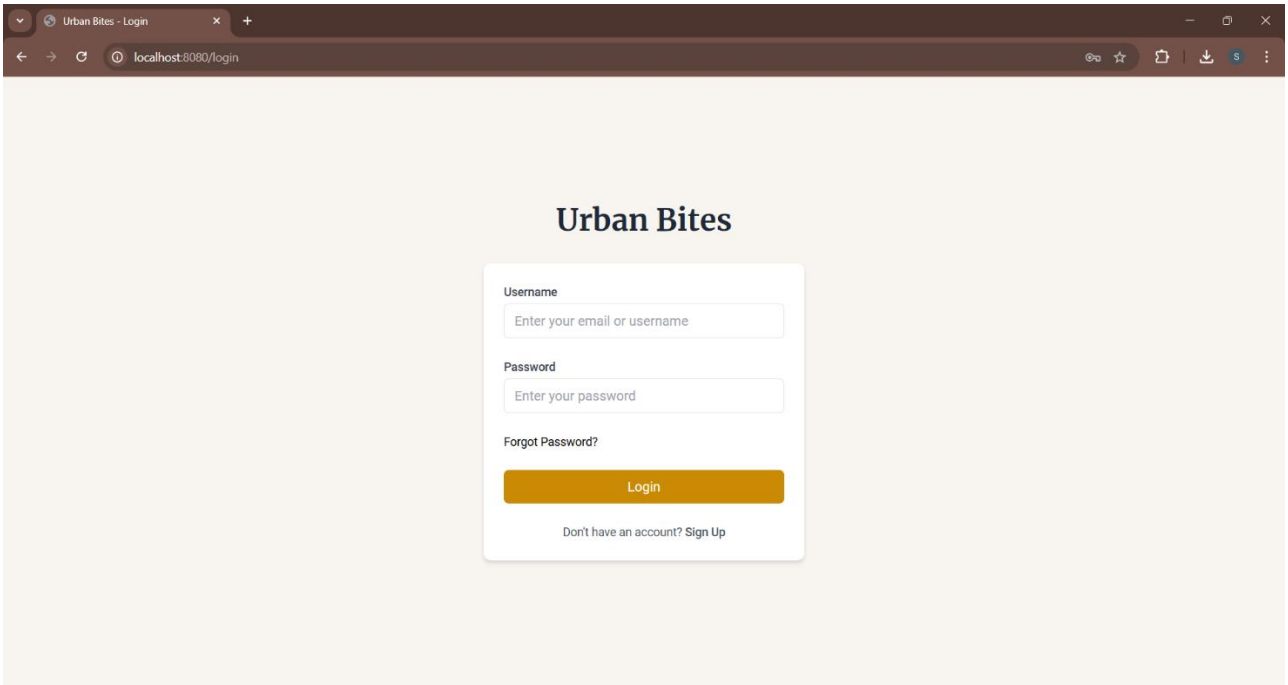


Fig. 5.1 The Login Page

The login page also ensures accessibility by being fully responsive and optimized for various devices, from desktops to smartphones. Its intuitive design minimizes friction, ensuring that users can quickly access the main page and begin exploring the application's features without unnecessary delays. This focus on secure and convenient access enhances overall user satisfaction and builds trust in the application.

Main Page The main page of the website serves as an inviting entry point for users. It features a visually striking layout that combines aesthetics with functionality. High-resolution photographs of the restaurant's ambiance and signature dishes create a lasting first impression, enticing users to explore further. These images are carefully curated to highlight the unique atmosphere of the

restaurant and showcase its culinary expertise.

In addition to its visual appeal, the main page provides essential information at a glance. The restaurant's operational hours are prominently displayed in a clear and readable format, ensuring users are well-informed about when the establishment is open. A dynamic banner or carousel may be used to highlight special promotions, featured dishes, or upcoming events, keeping the content fresh and engaging for returning visitors.

Navigation is seamlessly integrated into the main page, with intuitive menus and buttons guiding users to key sections such as the reservation page, menu, and home delivery options. Quick access links and call-to-action buttons are strategically placed to enhance usability and reduce the time users spend searching for relevant information. The main page is fully responsive, ensuring that it delivers a consistent and enjoyable experience across devices, from desktops to smartphones.
Moreover, the main page may include a "What's New" section or customer testimonials to foster a sense of community and trust. By combining aesthetics, functionality, and relevant content, the main page sets the tone for the entire application, creating an inviting and user-friendly platform for all visitors.

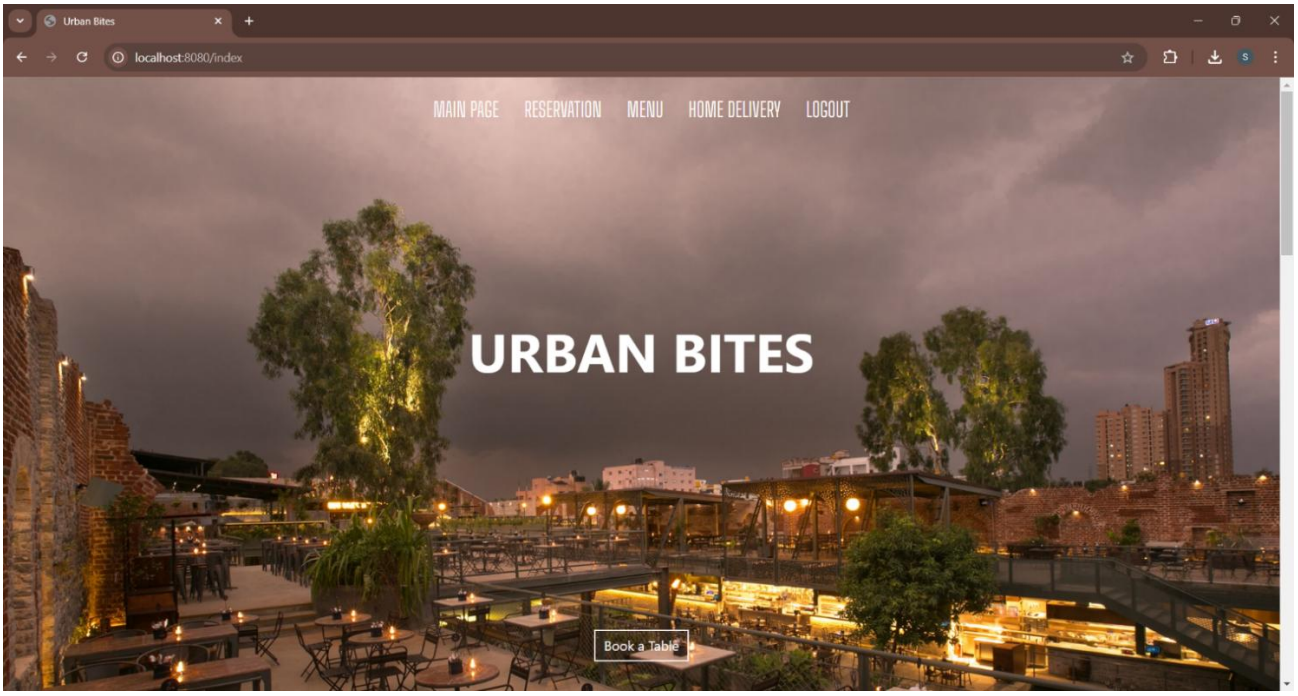


Fig 5.2 The Main Page

Reservation Page The reservation page is designed to simplify the table-booking process. Users can input their details, such as the desired date, time, and number of guests, to reserve a table effortlessly. The page features a clean and intuitive form with dropdown menus for selecting dates and times, ensuring a smooth user experience. Real-time availability checks prevent double bookings, providing users with immediate feedback if their chosen slot is unavailable.

To enhance user convenience, the reservation page also allows customers to include special requests,

such as seating preferences or dietary requirements. Once the reservation is submitted, users receive a confirmation email or notification summarizing their booking details, providing reassurance and clarity.

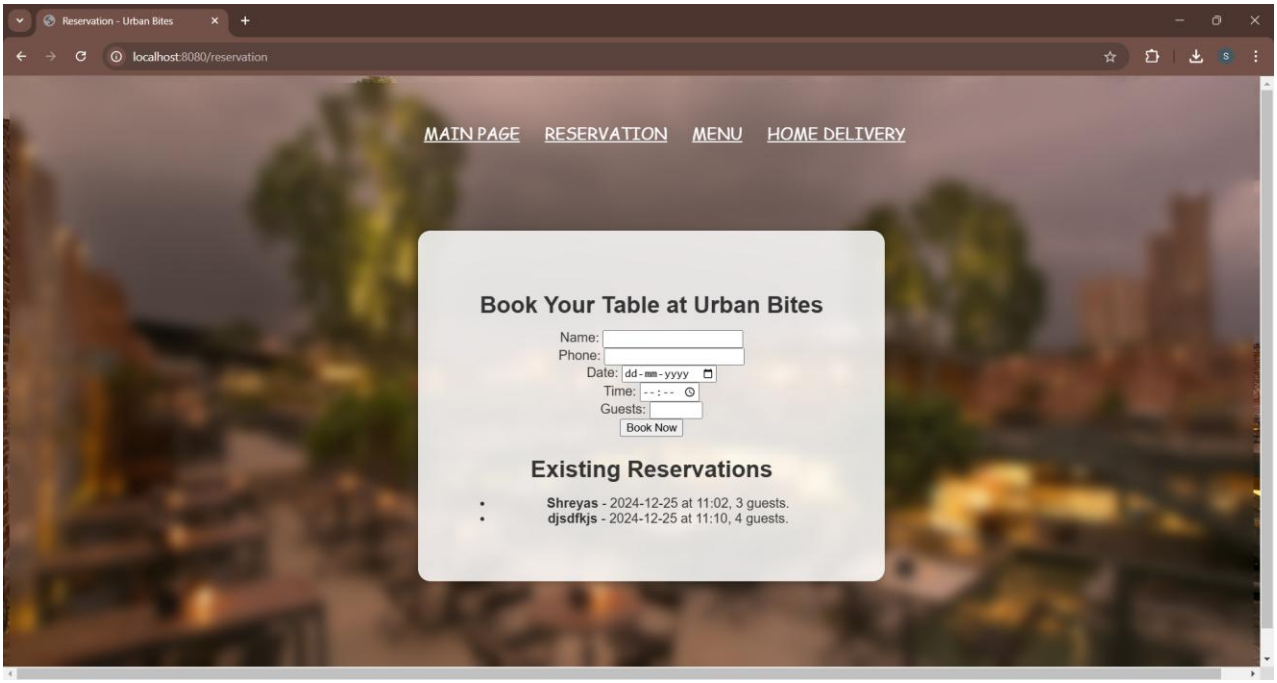


Fig. 5.3 The Reservation Page

The page is optimized for all devices, allowing users to make reservations on the go. Whether accessed via a smartphone, tablet, or desktop, the reservation page ensures a consistent and efficient experience. By eliminating the need for phone calls and reducing wait times, this feature offers a hassle-free solution for securing a table at the restaurant.

Menu Page The menu page showcases the restaurant's diverse offerings in an organized and visually appealing manner, serving as a comprehensive guide for customers to explore the culinary options. Users can browse through categorized sections, such as appetizers, main courses, desserts, and beverages, ensuring that the menu is easy to navigate. Each dish is accompanied by high-quality images, detailed descriptions, and key information such as ingredients, portion sizes, and pricing, allowing users to make informed choices based on their preferences and dietary requirements.

Interactive features enhance the user experience, including filters to sort dishes by cuisine type, popularity, or dietary restrictions like vegetarian, vegan, or gluten-free options. The page may also display chef's recommendations and customer ratings for each item, providing additional insights and guiding users toward popular choices.

For returning customers, a personalized section highlights previously ordered items or favorites, streamlining the decision-making process. Additionally, seasonal, or limited-time offerings are prominently displayed to encourage users to try new dishes. The menu page is fully responsive, ensuring seamless usability across devices, from smartphones to desktop computers, and maintains a visually cohesive and intuitive design.

Overall, the menu page is designed not only to inform but also to engage and delight users, enhancing their overall dining experience with the restaurant.

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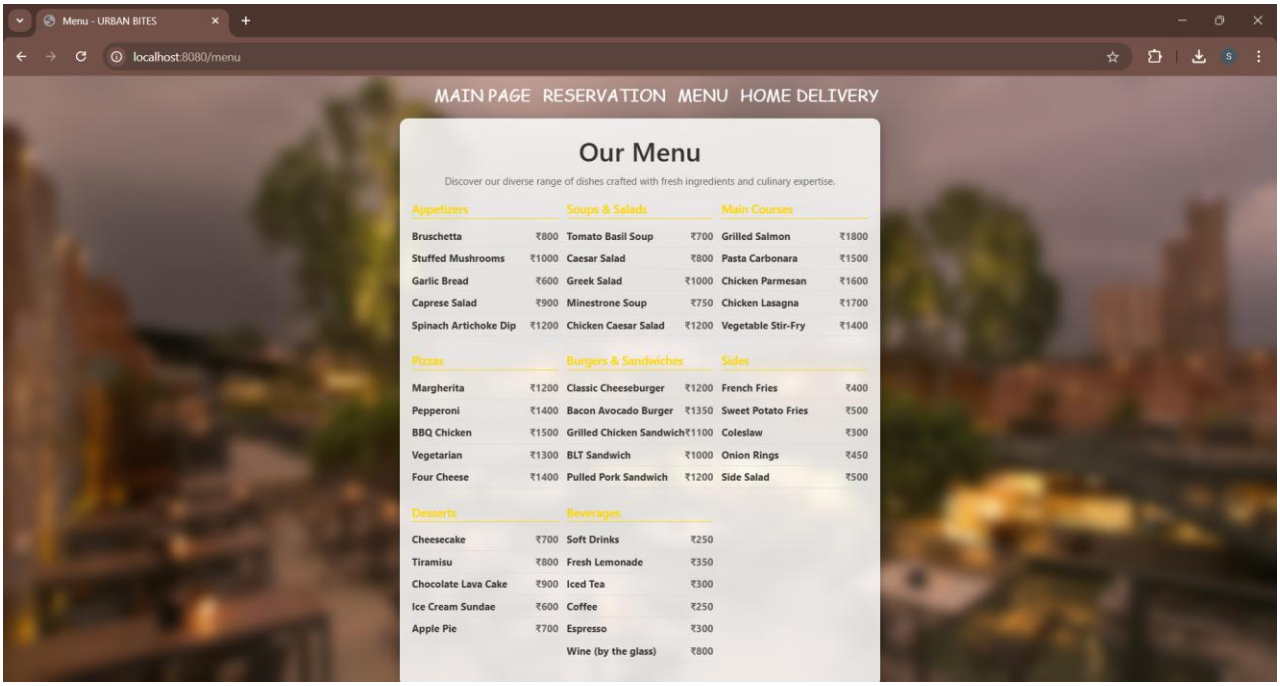

Fig 5.4 The Menu Page

Home Delivery Page The home delivery page is tailored to meet the needs of customers who prefer dining at home. Users can input their address details, including street name, city, and postal code, ensuring accurate delivery. A dropdown menu or autocomplete feature simplifies address entry and reduces errors.

The page allows users to browse and select food items for delivery from the menu, with each selection dynamically updating the total bill displayed at the bottom of the page. A checkbox feature provides a user-friendly way to finalize the order, ensuring clarity and precision in the selection process. Additionally, a summary section highlights the chosen items, quantities, and individual prices, helping users review their order before confirmation.

For added convenience, a corner of the page displays pending deliveries or order details in real-time, allowing users to track the status of their orders. Notifications or alerts can inform users about estimated delivery times or any changes to their order status. The integration of a secure payment gateway ensures a seamless transaction process.

The page is designed to be fully responsive, offering a consistent experience across all devices, including smartphones, tablets, and desktops. Its intuitive layout and visual elements prioritize user comfort, enhancing the overall convenience and efficiency of the home dining experience. This well-structured and interactive design ensures that the home delivery process is smooth, transparent, and user-friendly.

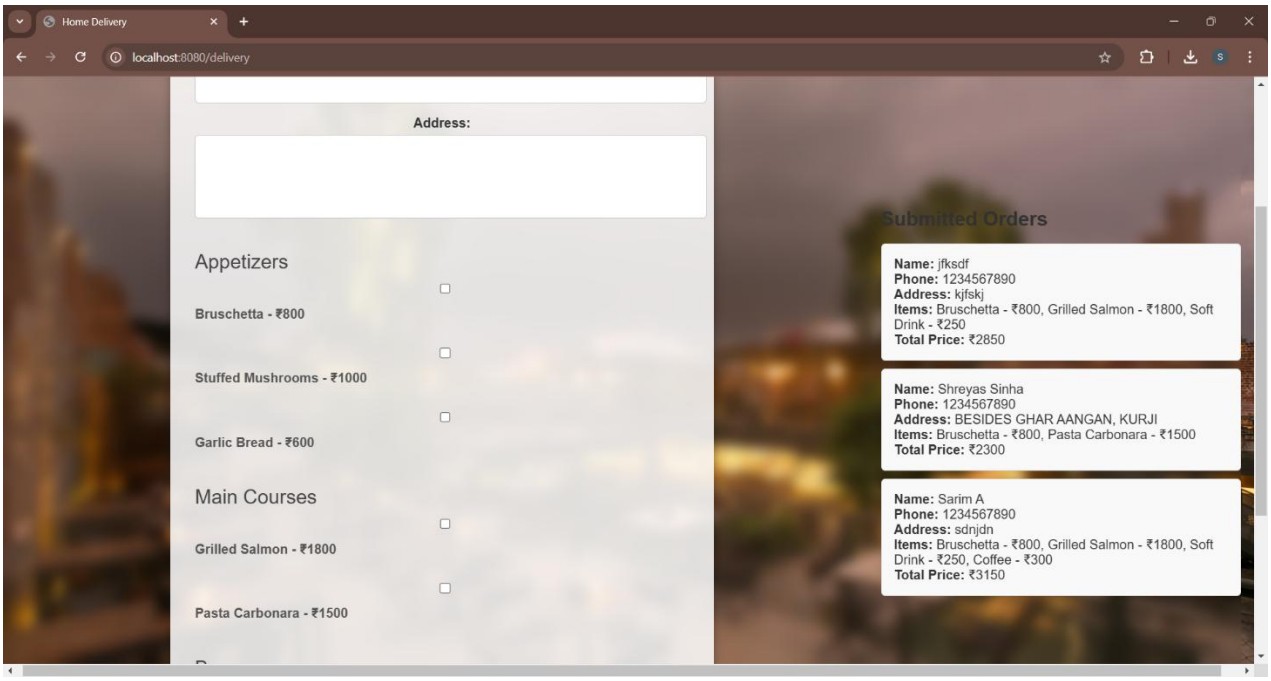

Fig 5.5 The Home Delivery Page

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